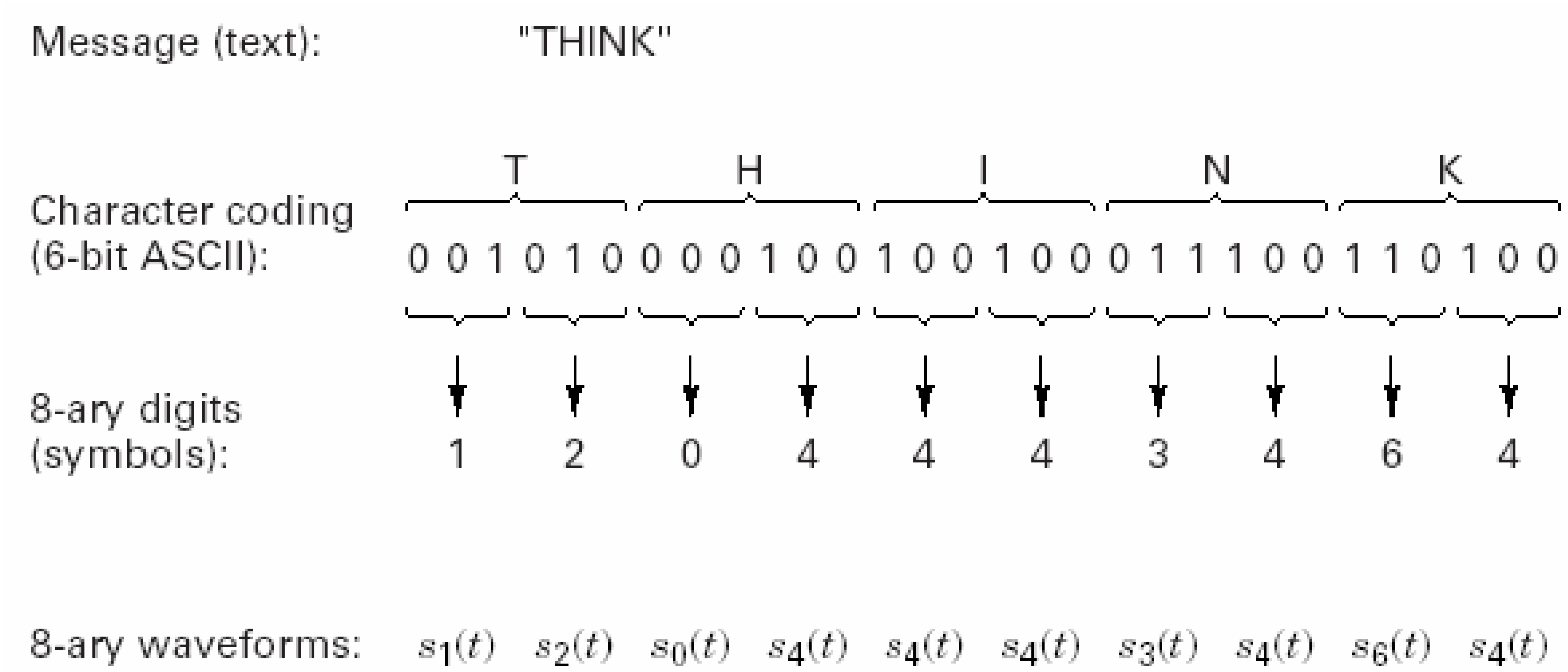
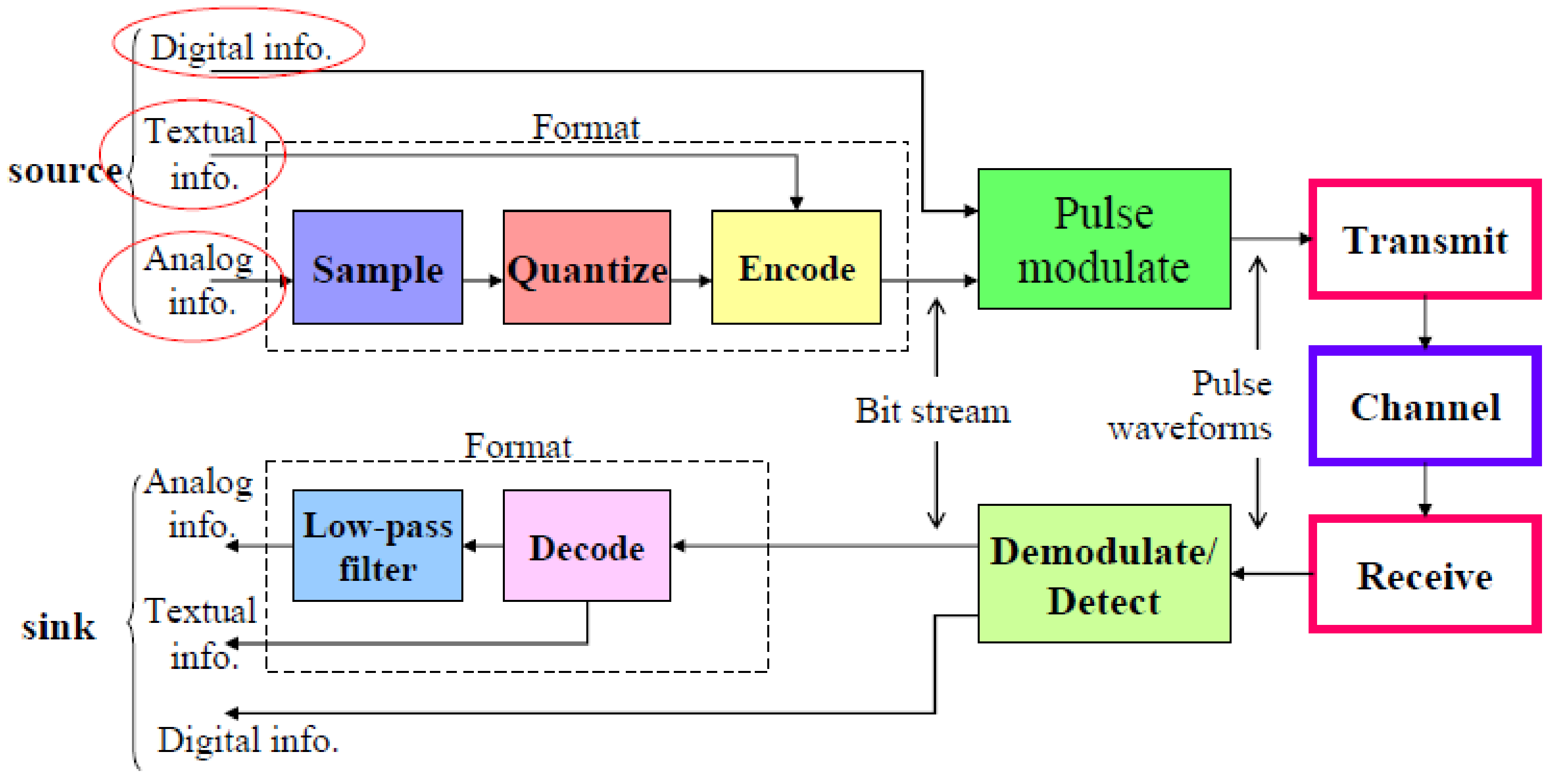
Lecture 2: Pulse Modulation

Introduction to Pulse Modulation

- Information signal is being transmitted in pulses
- Some parameter of a pulse train is varied in accordance with the message signal
- Pulse modulation includes many different methods of converting information into pulse form for transferring pulses from a source to a destination.



Message/Character Coding/Symbol



Formatting and transmission of base band signal

Category of PM

- Divided into two categories;
  1. Analog Pulse Modulation (APM)
  2. Digital Pulse Modulation (DPM)

PULSE MODULATION

- Some parameter of a pulse train is varied in accordance with the message signal

I. **Analog Pulse Modulation** – A periodic pulse train is used as a career wave, and some characteristic feature of each pulse (e.g. Amplitude, Duration or Position) is varied in a continuous manner in accordance with the corresponding sample value of the message signal. Basically, information is transmitted in analog form, but the transmission takes place at discrete times

II. **Digital Pulse Modulation** - The message signal is represented in a form that is discrete in both time and amplitude, thereby permitting its translation in digital form as a sequence of coded pulses

PULSE MODULATION

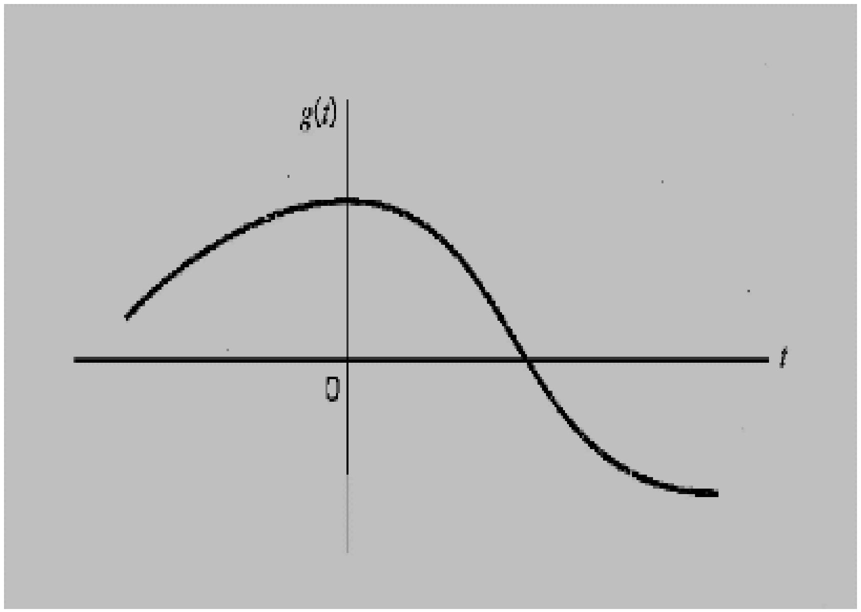
- **It involves sampling, quantization and encoding**
- **Converting samples into discrete pulses**
- **Transport the pulses over the physical transmission medium.**
- **Four (4) Methods**

1. PAM		
2. PWM	**Analog Pulse Modulation**	
3. PPM		
4. PCM	**Digital Pulse Modulation**	

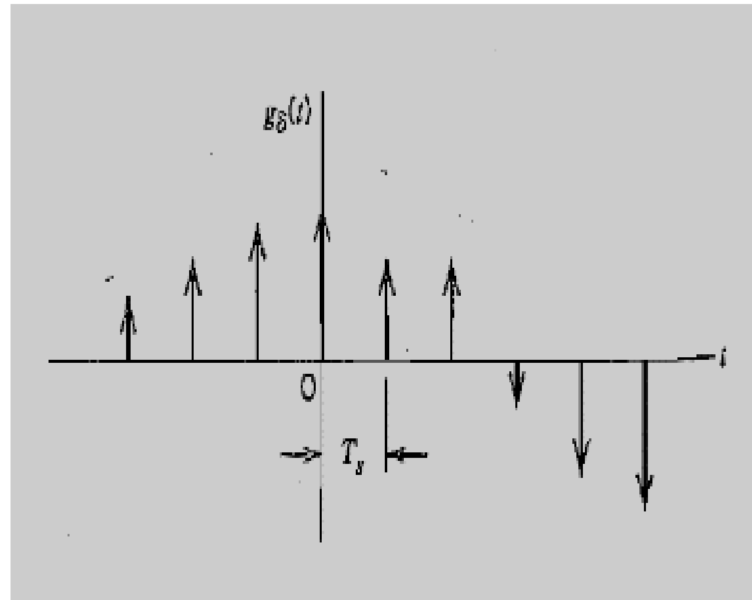
PULSE MODULATION :Sampling

- What is sampling?
- Sampling is the process of taking periodic sample of the waveform to be transmitted.
- "the more samples that are taken, the more final outcome looks like the original wave.
- However if fewer samples are taken, then other kinds information could be transmitted."

SAMPLING Ctd…



Analog Signal



Instantaneously sampled version of analog signal

This signal is sampled instantaneously and at a uniform Rate, once every T_s Seconds.

T_S = Sampling Period
$f_s = 1/T_s$ = Sampling Rate

This process of uniformly sampling a continuous-time signal of finite energy results in a periodic spectrum with a Period equal to the sampling rate.

Sampling Theorem

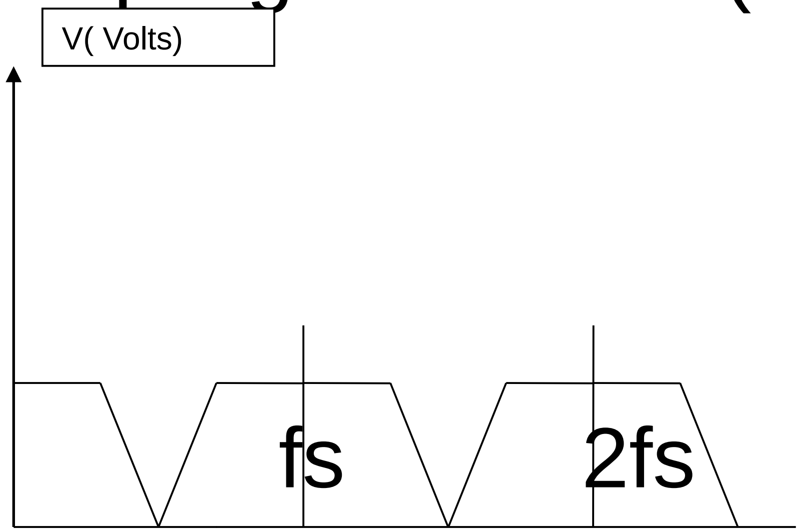
Sampling theorem (Nyquist's theorem)

- - is used to determine minimum sampling rate for any signal so that the signal will be correctly restored at the receiver.
- Nyquist's theorem states that,
- "The original information signal can be reconstructed at the receiver with minimal distortion if the sampling rate in the pulse modulation system is equal to or greater than twice the maximum information signal frequency."

-
- sampling frequency
- $$f_s \geq 2 f_m$$
- fs=sampling frequency and
- fm(max) = maximun frequency of the modulating signal.

Condition of sampling process

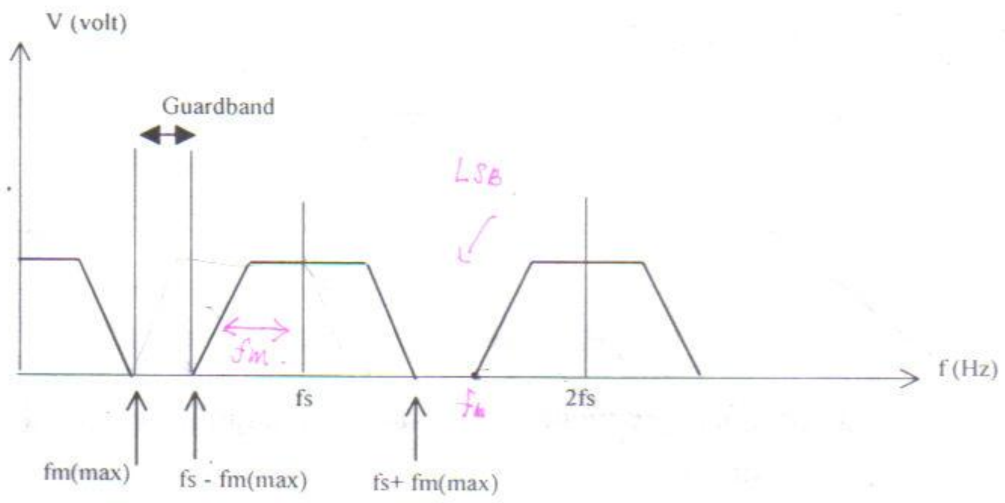
- 1) sampling at Fs =2fm(max)



- figure 2.1 :
- Frequency spectrum of modulating signal when sampled at fs=2fm(max)

- When the modulating is sampled at a minimum sampling frequency, the frequency spectrum is as shown in figure 2.1.
- In practice it is difficult to design a low pass filter, in order to restore the original modulating signal

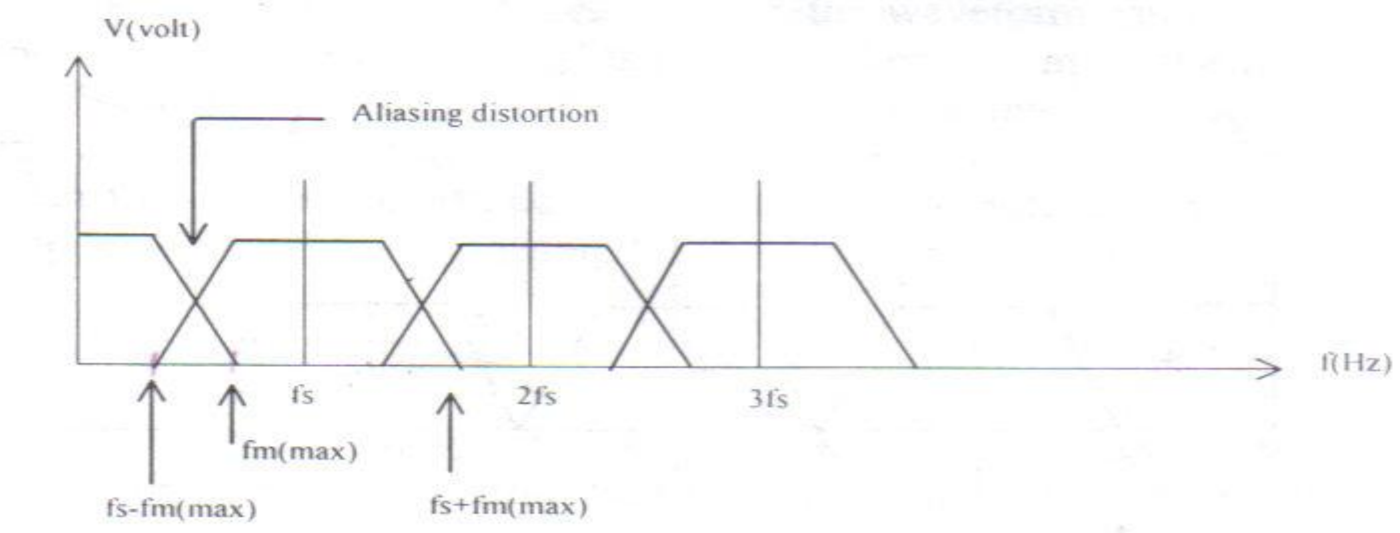
- 2) Sampling at fs> 2fm(max)
- This sampling rate creates a guard band between fm(max) and the lowest frequency component (fs-fm(max)) of the sampling harmonics.
- Therefore a more practical LPF can be used to restore the modulating signal.



- Figure 2.2 Sampling at fs> 2fm(max)

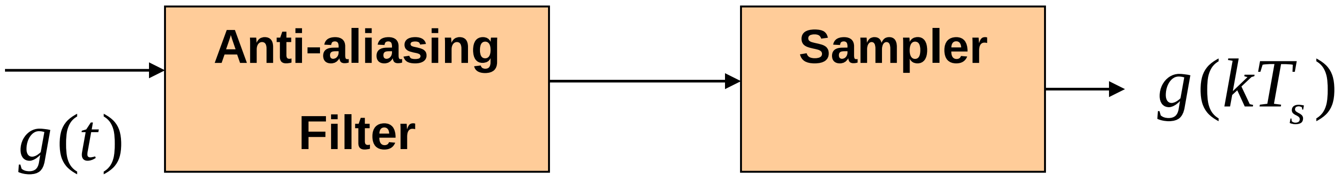
3) Sampling at fs < 2fm(max)

- When the sampling rate is less than the minimum value, distortion will occurs. This distortion is called aliasing.



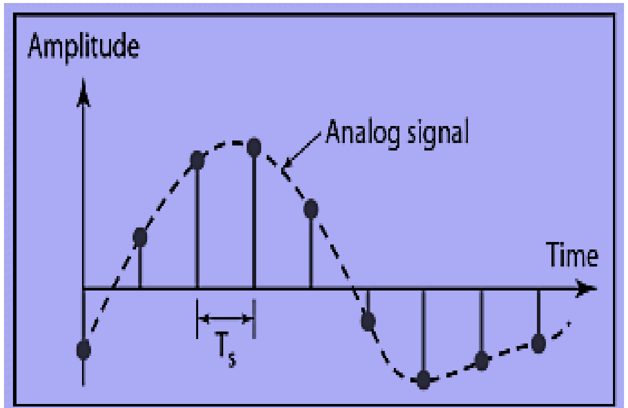
- Figure 2.3 Sampling at fs < 2fm(max)

- **Aliasing** is a phenomenon of a high frequency component in the spectrum of the signal seemingly to take in the identity of a lower frequency in the spectrum of its sampled version
- Aliasing effect can be eliminated by using an anti-aliasing filter prior to sampling and using a sampling rate slightly higher than Nyquist rate (*fs*=2*W*).

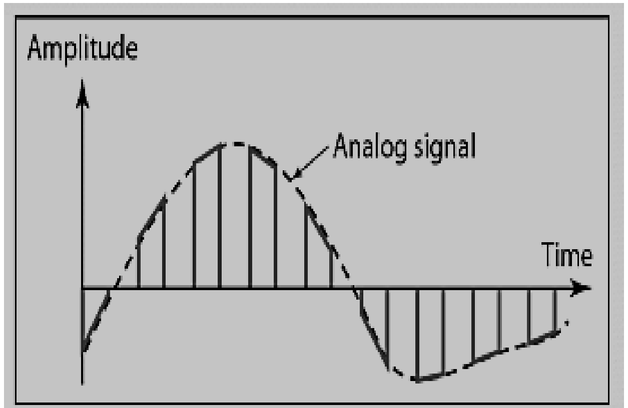


Types of sampling methods

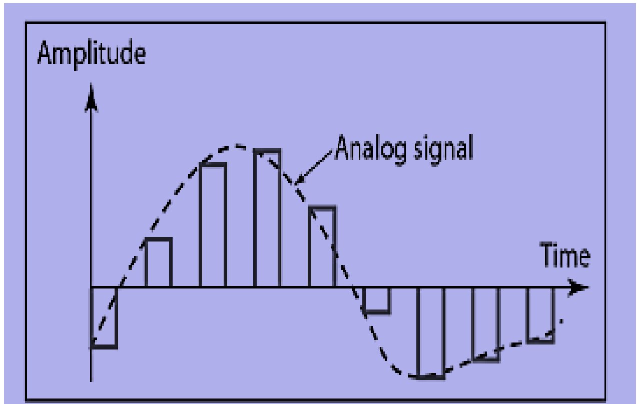
- There are three sampling methods.



a. Ideal sampling



b. Natural sampling



c. Flat-top sampling

Ideal Sampling

- An Impulse is taken at each sampling Instant

Natural Sampling

- A pulse of short width with varying Amplitude

Flat-top Sampling

- Sample and hold like natural sampling but with single Amplitude value.

ANALOG PULSE MODULATION (APM)

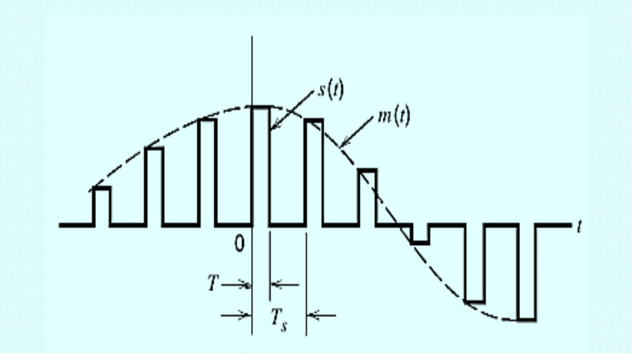
- In APM, the carrier signal is in the form of pulse waveform, and the modulated signal is where one of the characteristic (either amplitude, width or position) is changed according to the modulating/audio signal
- The three common techniques of APM are: Pulse Amplitude Modulation (PAM),

  Pulse Width Modulation (PWM) and

  Pulse Position Modulation (PPM).

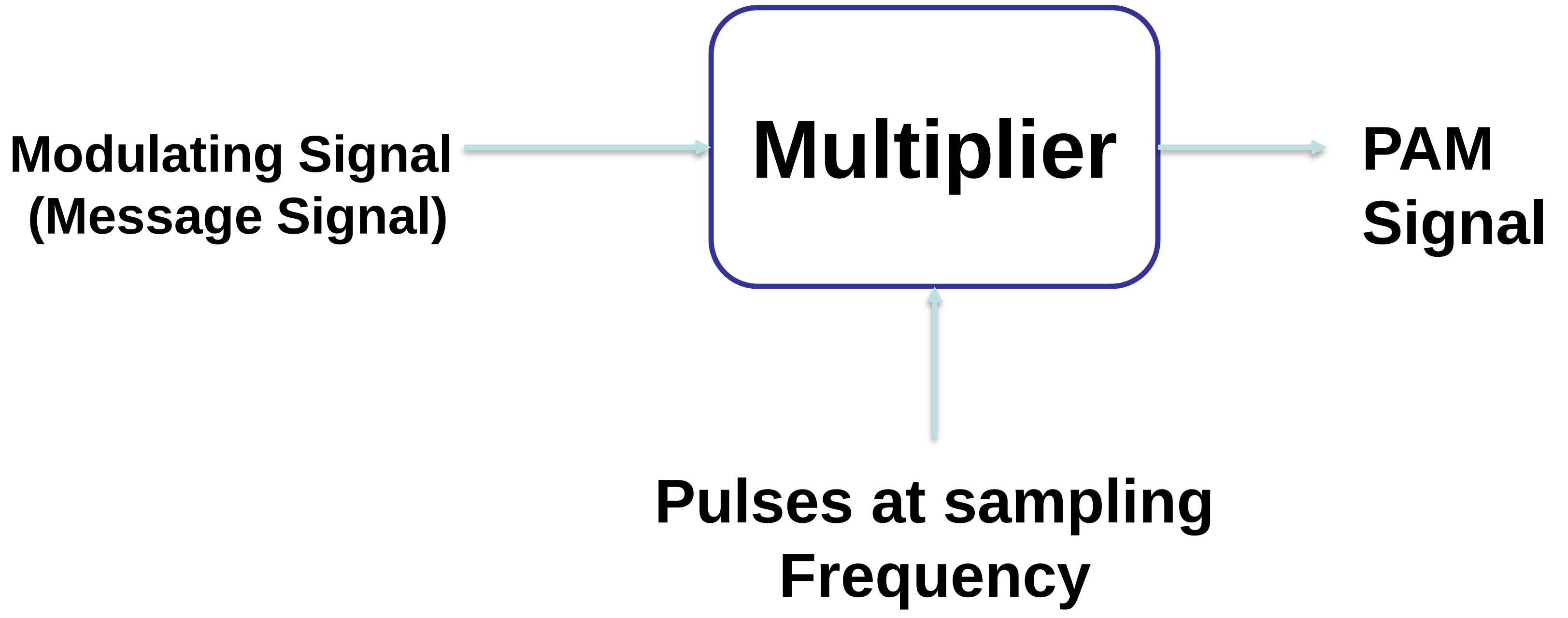
Pulse Amplitude Modulation (PAM)

- The simplest form of pulse modulation
- The amplitude of a constant width, constant position pulse (carrier signal) is varied according to the amplitude of the modulating signal.
- Basically the modulating signal is sampled by the digital train of pulses and the process is based upon the sampling theorem



PAM Ctd…

- The carrier is in the form of narrow pulses having frequency fs. The uniform sampling takes place in multiplier to generate PAM signal. Samples are placed Ts sec away from each other.



PAM Ctd…

- Depending upon the shape and polarity of the sampled pulses, PAM is of two types

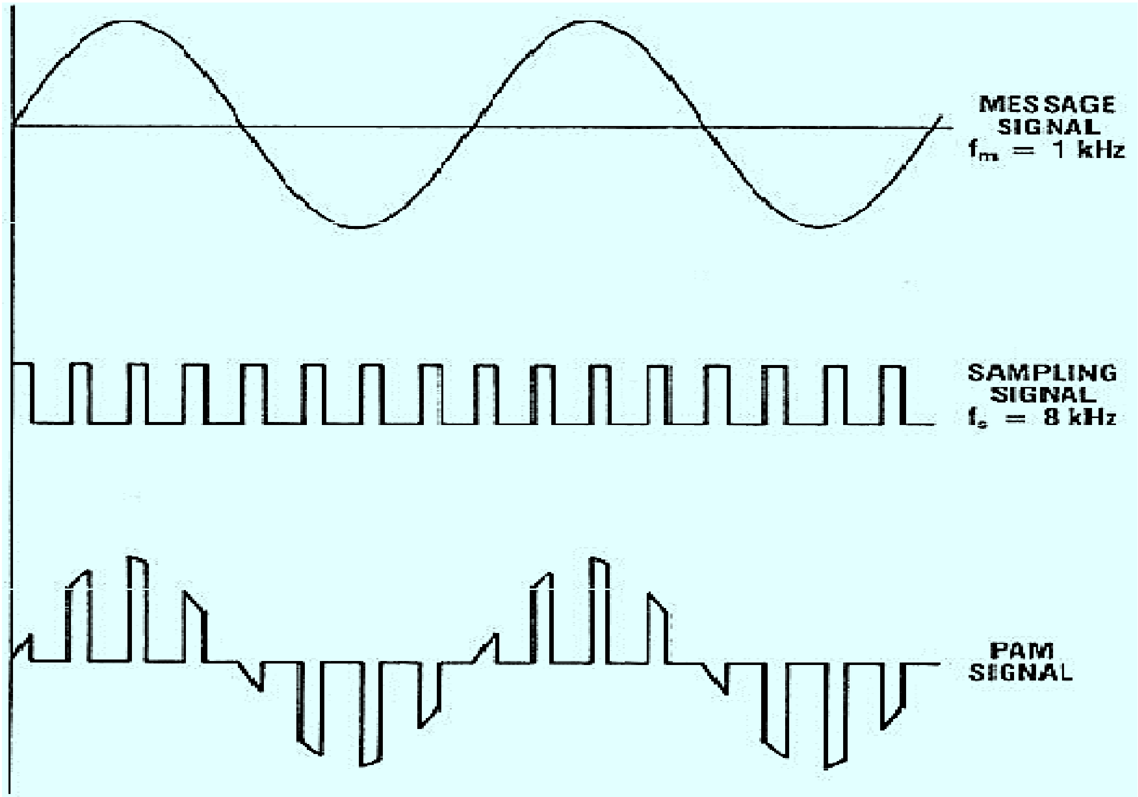
i. Natural PAM

ii. Flat-topped PAM

PAM Ctd…

Natural PAM

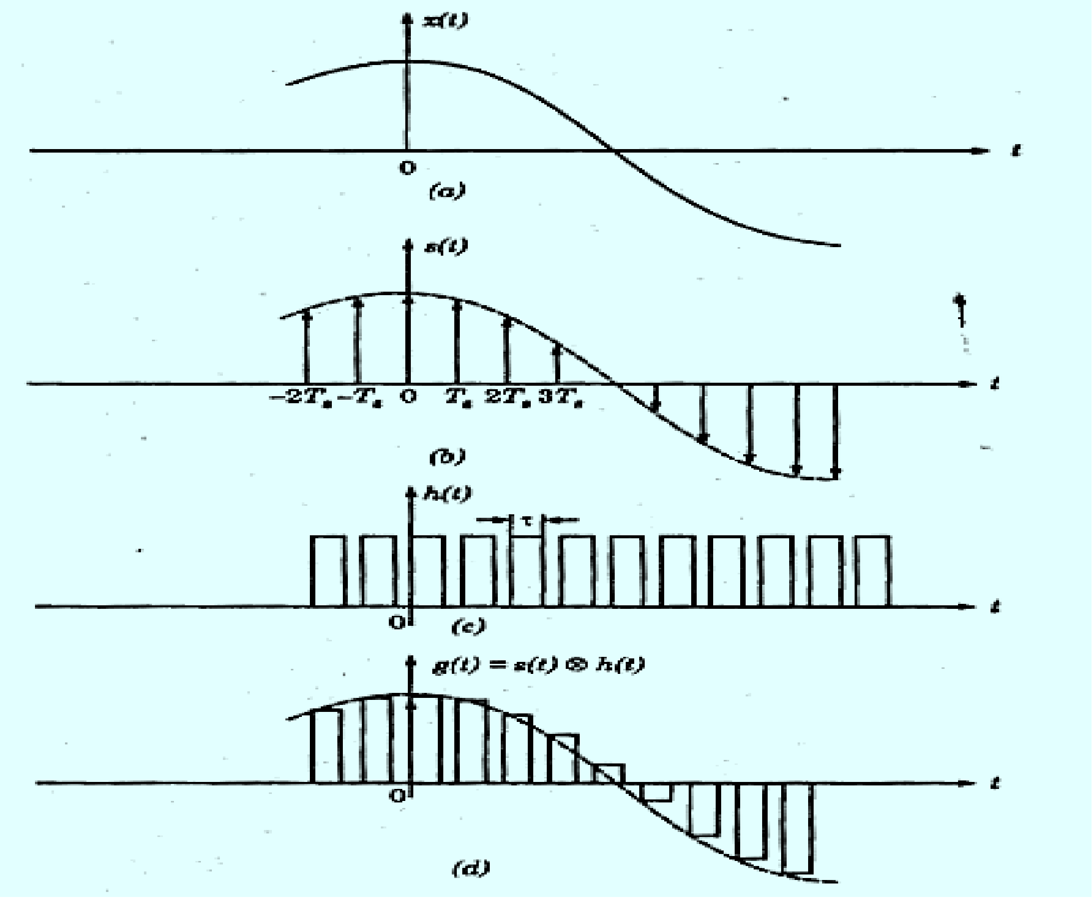
sampling occurs when top portion of the pulses are subjected to follow the modulating signal



PAM Ctd…

Flat-Topped PAM

pulses have flat tops after modulation.



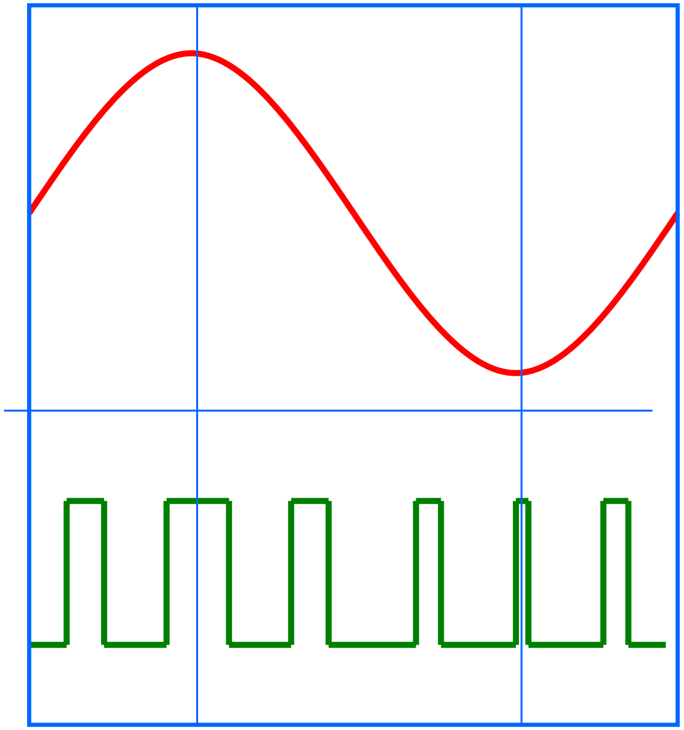
Pulse Width Modulation (PWM)

- The technique of varying the width of the constant amplitude pulse proportional to the amplitude of the modulation signal.
- Also known as Pulse Duration Modulation (FDM).
- Either the leading edge, trailing edge or both may be varied by the modulating signal.

Pulse Width Modulation (PWM)

- PWM gives better signal to noise performance than PAM.
- PWM has advantage, when compared with PPM, that is its pulse are of varying width and therefore of varying power content. PWM still works if synchronization between transmitter and receiver fails, whereas PPM does not.

PWM Ctd…

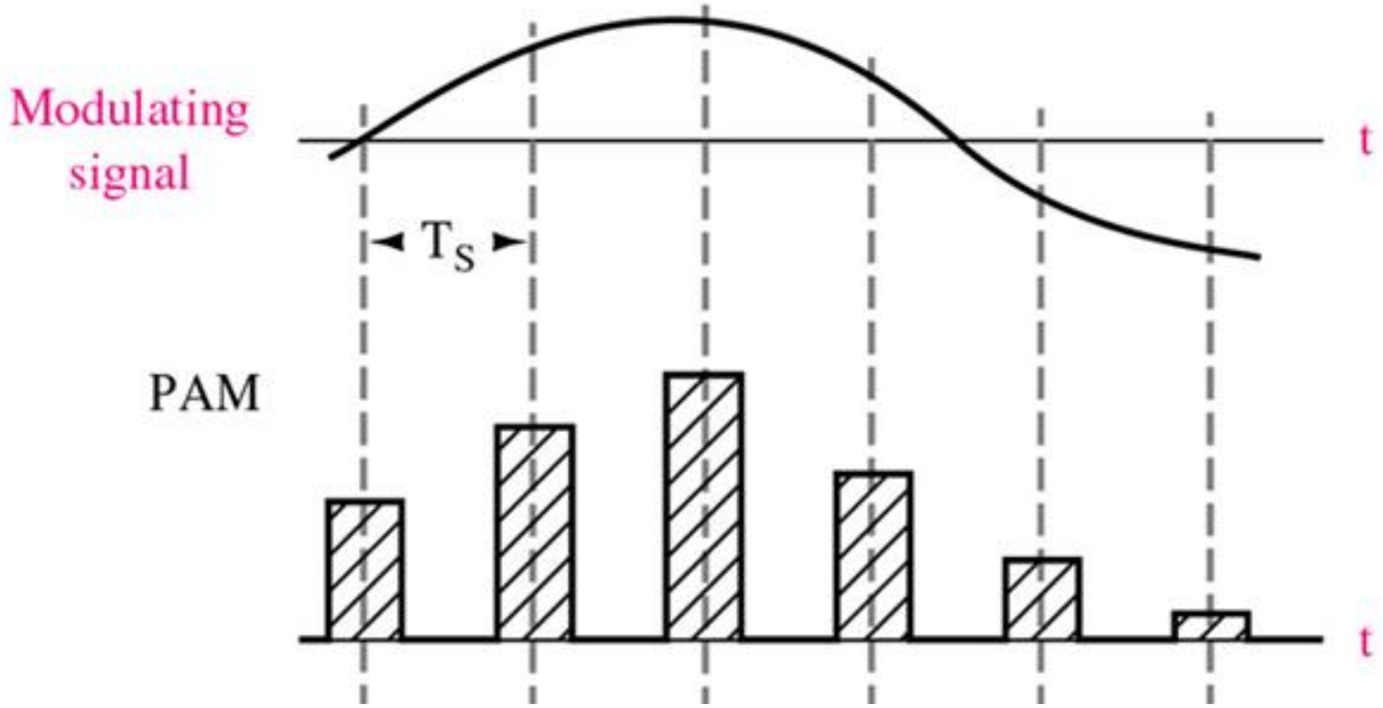


Analog Signal

Width Modulated Pulses

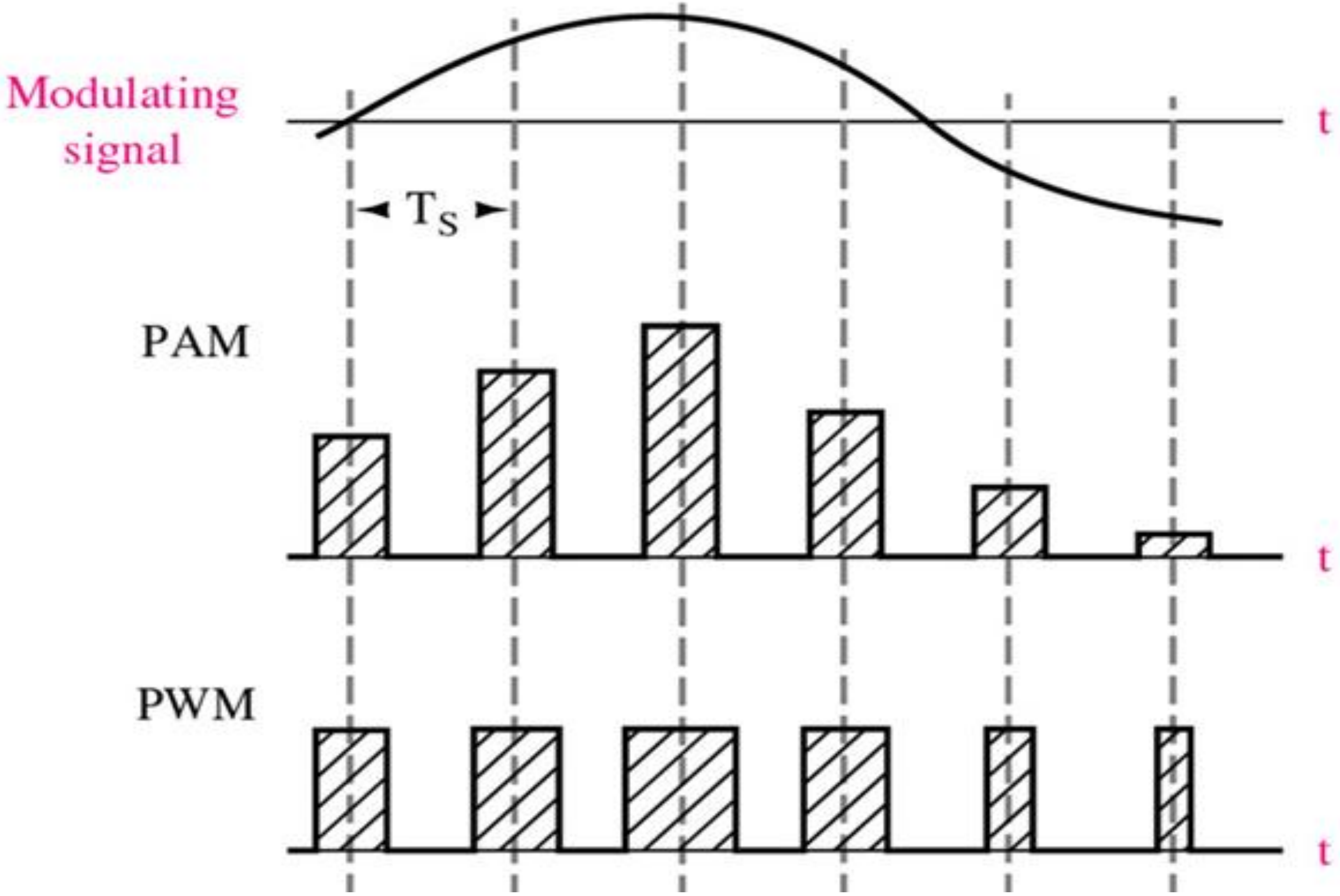
Pulse Position Modulation (PPM)

- PPM is when the position of a constant-width and constant-amplitude pulse within prescribed time slot is varied according to the amplitude of the modulating signal.
- PPM has the advantage of requiring constant transmitter power output, but the disavantage of depending on transmitter-receiver synchronization.
- PPM has less noise due to amplitude changes, becaused the received pulses may be clipped at the receiver, thus removing amplitudeschanges caused by noise.



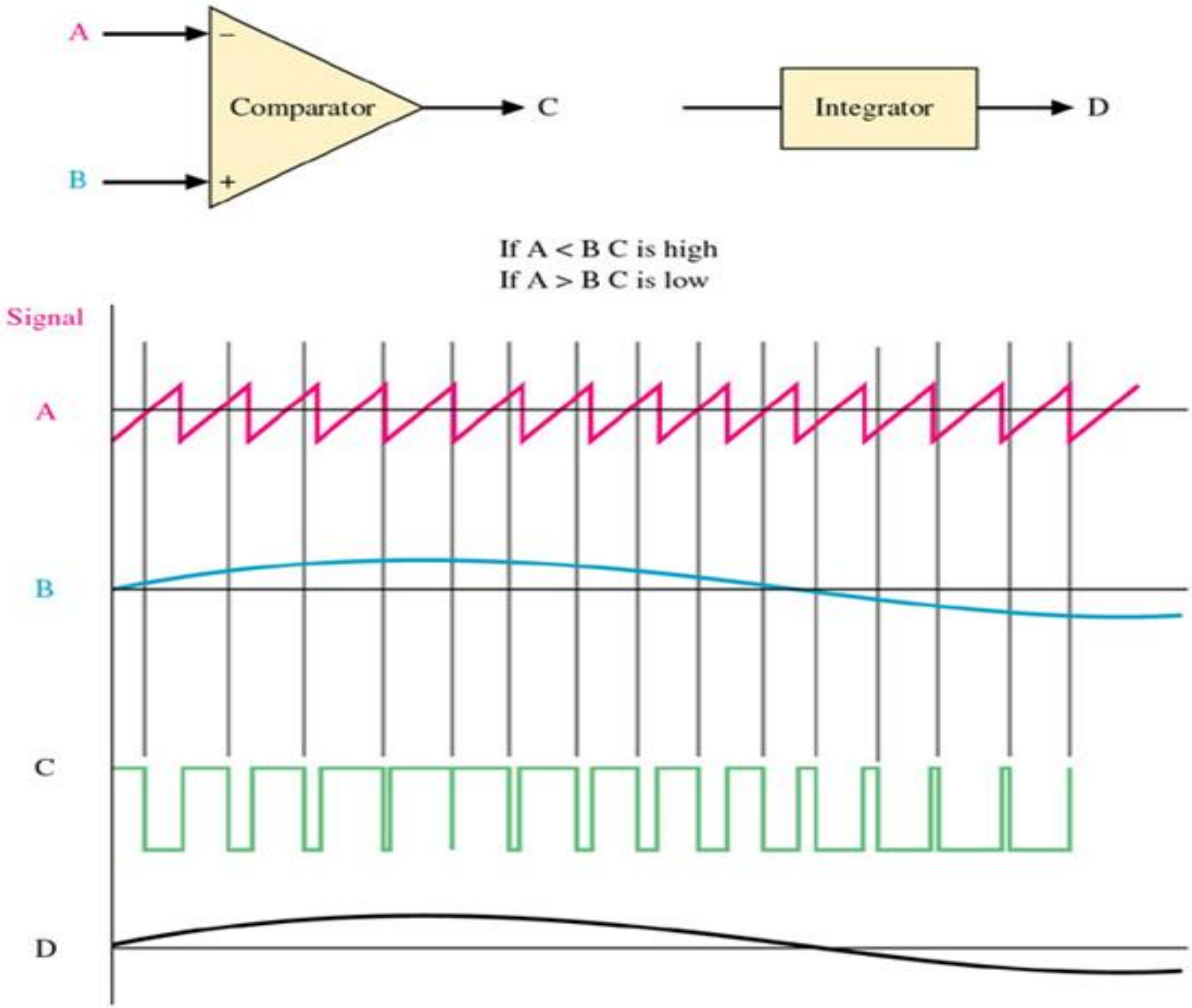
Pulse Amplitude Modulation (PAM)

Modulation in which the amplitude of pulses is varied in accordance with the modulating signal



Pulse Width Modulation (PWM)

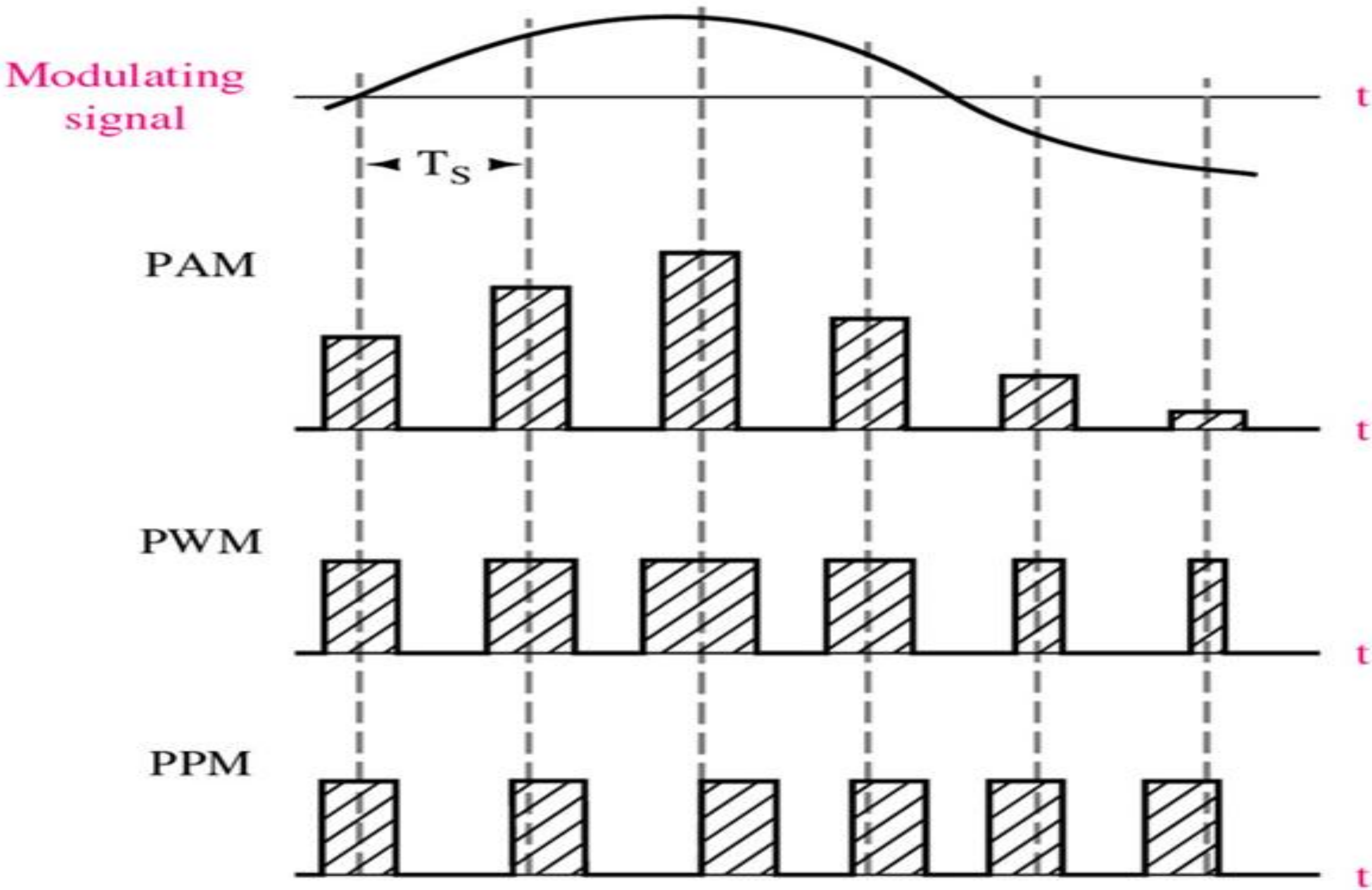
Modulation in which the duration of pulses is varied in accordance with the modulating signal



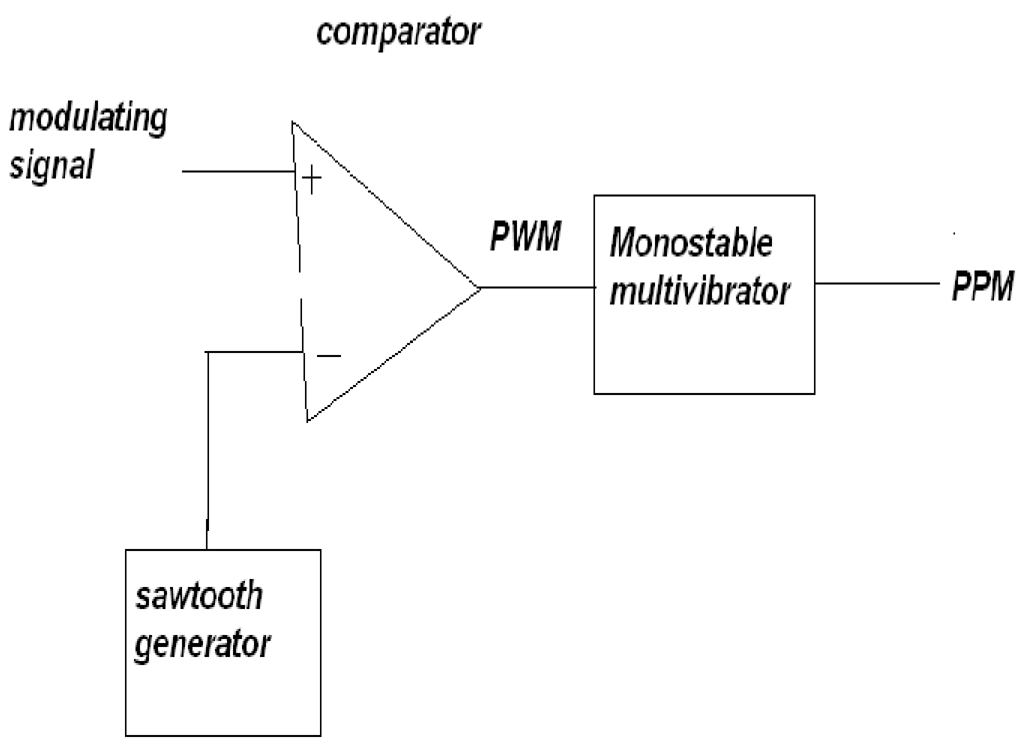
Pulse Width Modulation (PWM)

Pulse Position Modulation (PPM)

Modulation in which the temporal positions of the pulses are varied in accordance with some characteristic of the modulating signal.



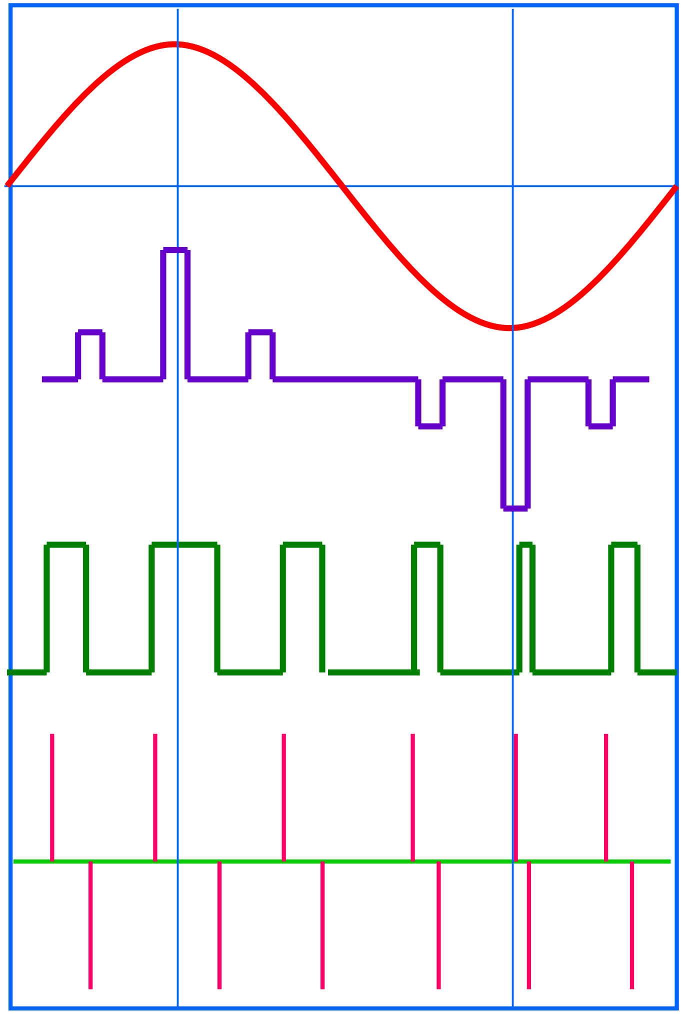
PWM, PPM GENERATION



PWM, PPM GENERATION Ctd…

- The PPM signal can be generated from PWM signal.
- The PWM pulses obtained at the comparator output are applied to a monostable multivibrator which is –ve edge triggered.
- Hence for each trailing edge of PWM signal, the monostable output goes high. It remains high for a fixed time decided by its own RC components

PAM, PWM and PPM



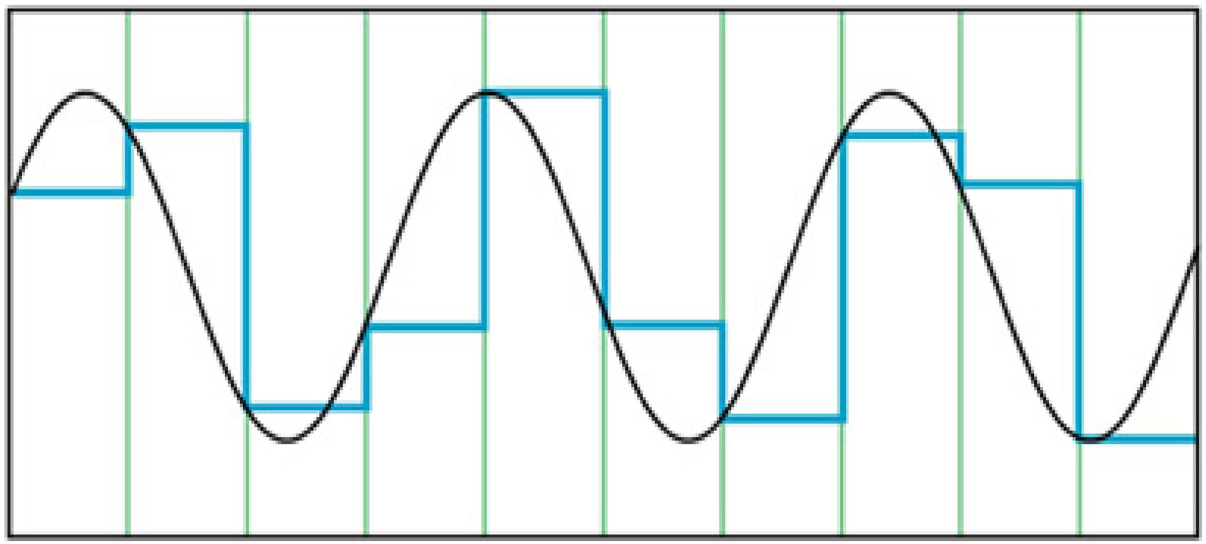
Analog Signal

Amplitude Modulated Pulses

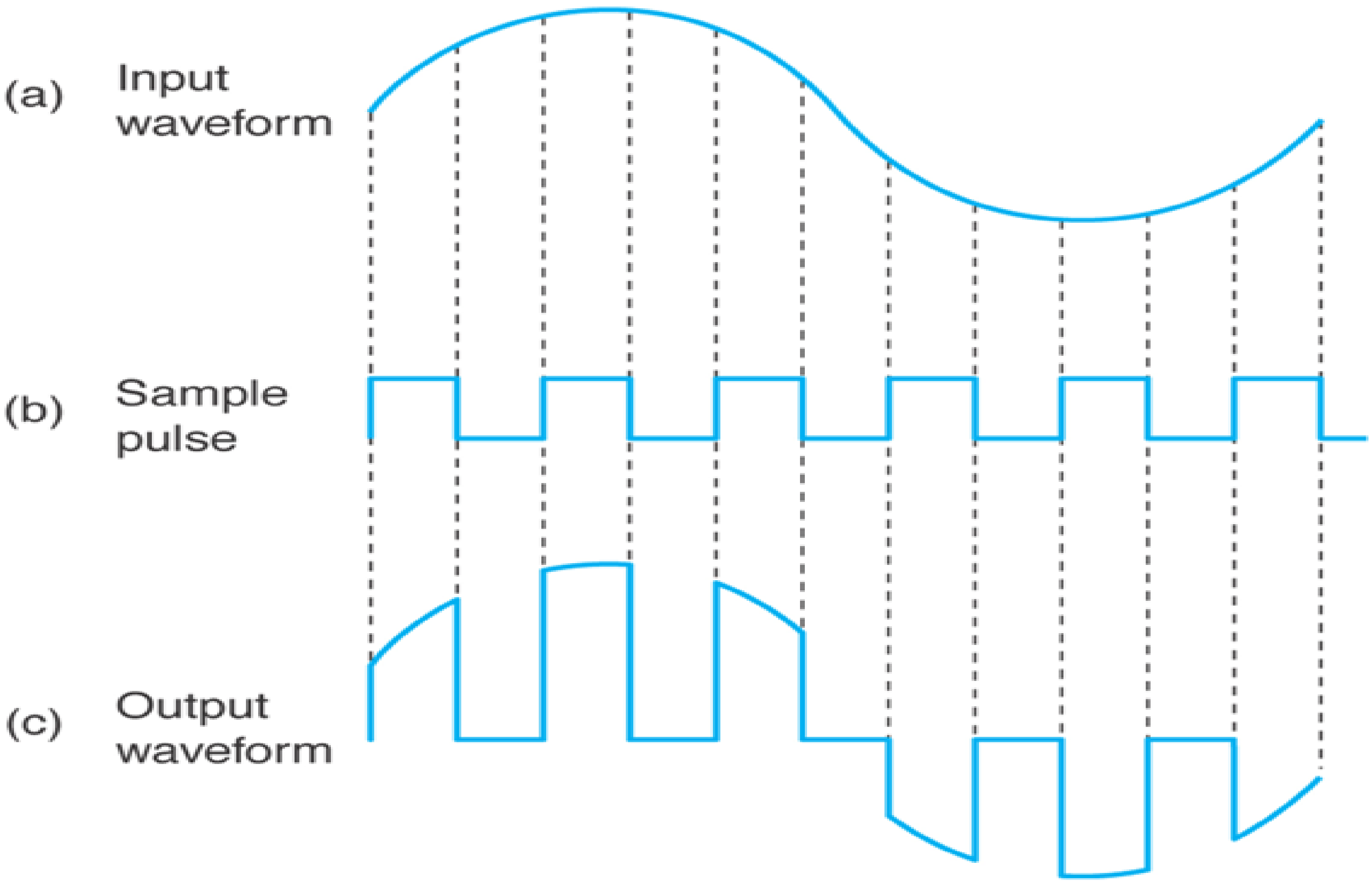
Width Modulated Pulses

Position Modulated Pulses

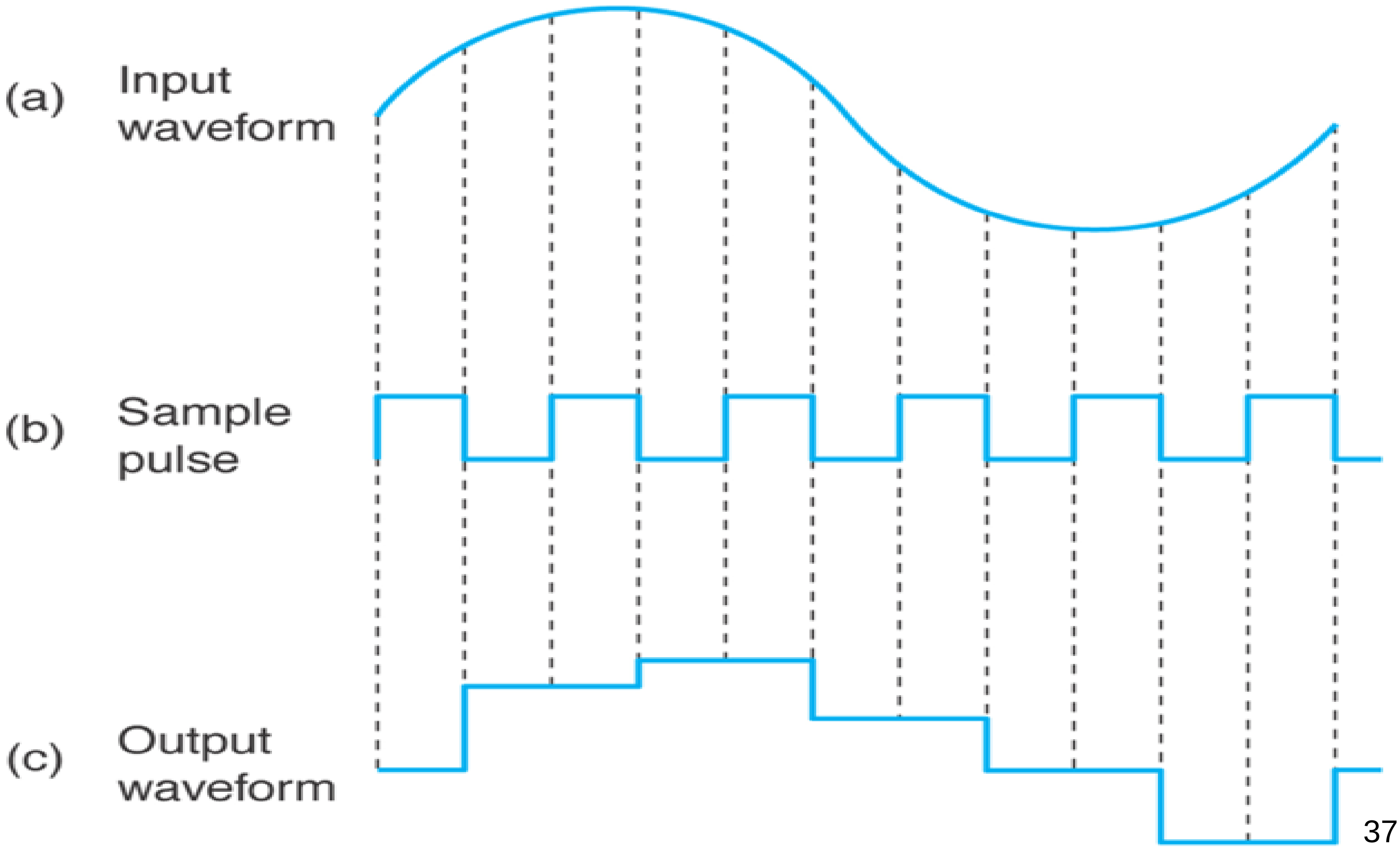
How to encode analog waveforms ?
(from analog sources into baseband digital signals)



Natural Sampling



Flat-top Sampling

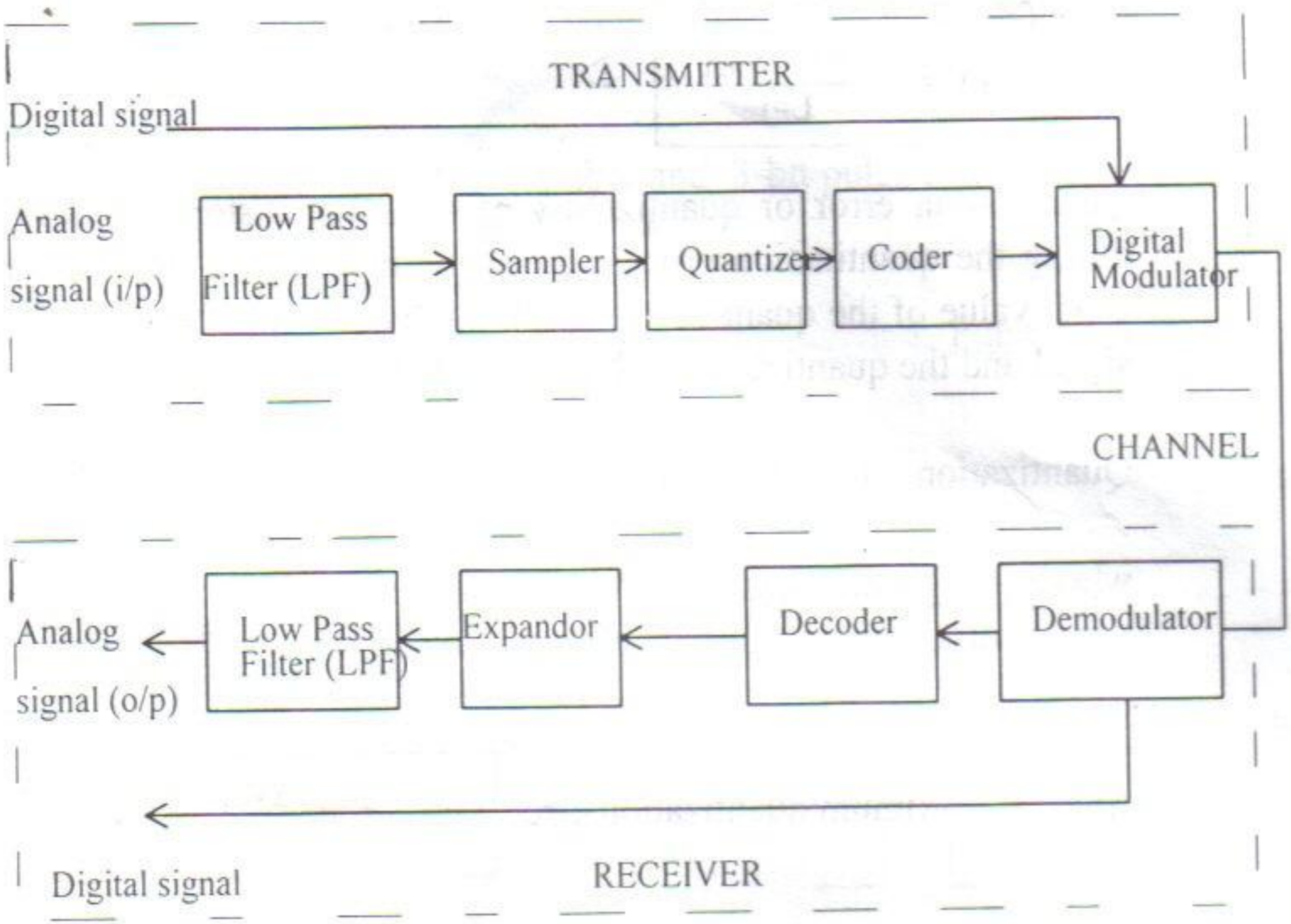


DIGITAL PULSE MODULATION (DPM)

Pulse Code Modulation (PCM)

- PCM is a form of digital modulation where group of coded pulses are used to represent the analog signal.
- The analog signal is sampled and converted to a fixed length, serial binary number for transmission.
- A block diagram of a PCM system is as shown in figure 2.4

Fig.2.4 A block diagram of PCM system (single channel)



Principles of PCM

- Three main process in PCM transmission are sampling, quantization and coding.
- 1. Sampling – is a process of taking samples of information signal at a rate of Nyquist's sampling frequency.
- 2. Quantization – is a process of assigning the analog signal samples to a pre-determined discrete levels. The number of quantization levels ,L, depends on the number of bits per sample, n, used to code the signal. Where $L = 2^n$

Principles of PCM (…cont.)

- The magnitude of the minimum stepsize of the quantization levels is called resolution, ΔV
- It is equal in magnitude to the voltage of the least significant bit of the magnitude stepsize of the digital to analog converter (DAC). The resolution depends on the maximum voltage, Vmax, and the minimum voltage Vmin of the information signal, where

$$\Delta V = \frac{V_{\max} - V_{\min}}{L-1}$$

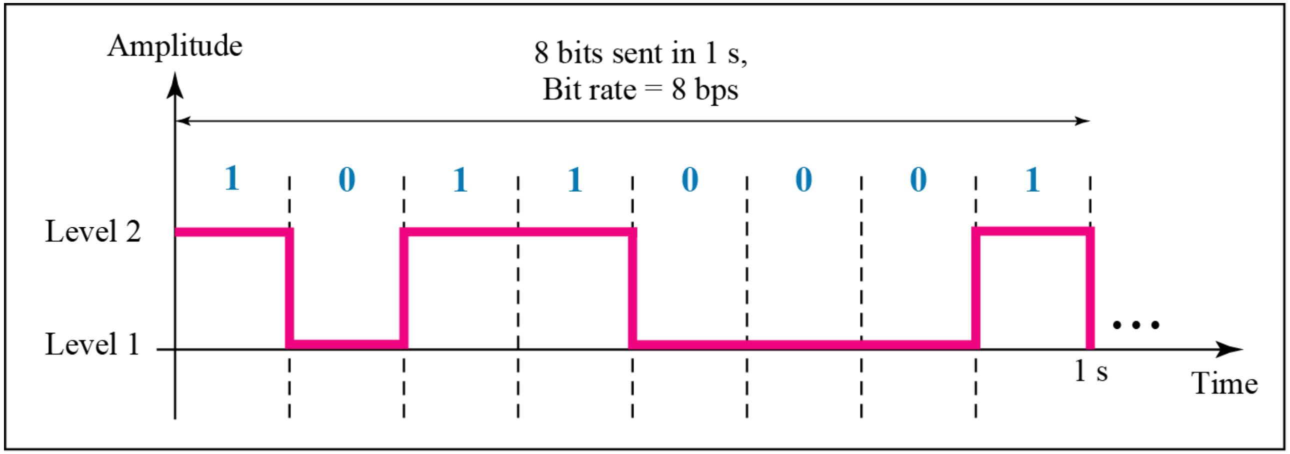
Principles of PCM (…cont.)

- Quantization error or quantization noise is the distortion introduced during the quantization process when the modulating signal is not an exact value of the quantized level. It is the difference between original signal and the quantized signal magnitude that is :
- Quantization error, Qe = |x(t)| - |q(t)|
- Where |x(t)| is the magnitude of original signal
- Where |q(t)| is the magnitude of quantized signal
- The maximum quantization error, $Qe_{\max} = \pm \frac{\Delta V}{2}$
- Quantization error can be reduced by increasing the number of quantization level BUT this will increase the bandwidth required.

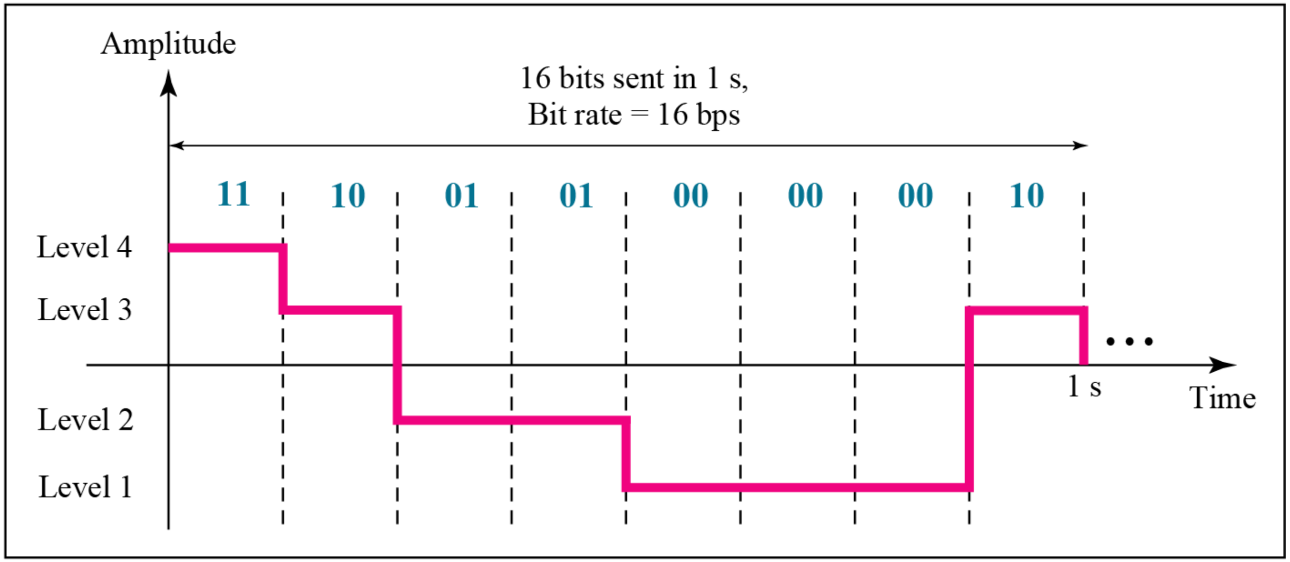
Principles of PCM (…cont.)

- ENCODING
- This is a process where each quantized sample is digitally encoded into n-bits codeword, where
- n = number of bits/sample $$n = \log_2 L$$
- L = number of quantization levels
- Transmission bit rate (R) is the rate of information transmission (bits/sec).
- It depends on the sampling frequency and the number of bit per sample used to encode the signal and is given by
- Transmission bit rate $$R = n \times f_s bits/\sec$$
- Transmission Bandwidth ; $$B = n \times f_s Hz$$

Two digital signals: one with two signal levels and the other with four signal levels



a. A digital signal with two levels



b. A digital signal with four levels

Examples

A digital signal has 8 levels. How many bits are needed per level?

We calculate the number of bits from the formula

$$\boxed{\text{Number of bits per level} = \log_2 8 = 3}$$

Each signal level is represented by 3 bits.

A digital signal has 9 levels. How many bits are needed per level?

Each signal level is represented by 3.17 bits.
The number of bits sent per level needs to be an integer as well as a power of 2.
Hence, 4 bits can represent one level.

Analog pulse vs Digital pulse modulation

Analog pulse modulation

- A periodic pulse train is used as the carrier wave
- Some characteristic feature of each pulse is varied in a continuous manner in accordance with the corresponding sample value of the message signal
- Analog pulse-modulation systems rely on the sampling process to maintain continuous amplitude representation of the message signal

Digital pulse modulation

- The message signal is represented in a form that is discrete in both time and amplitude
- Its transmission in digital form as a sequence of coded pulse
- Digital pulse-modulation system use not only the sampling process but also the quantization process.
- Digital modulation makes it possible to exploit the full power of digital signal-processing techniques.

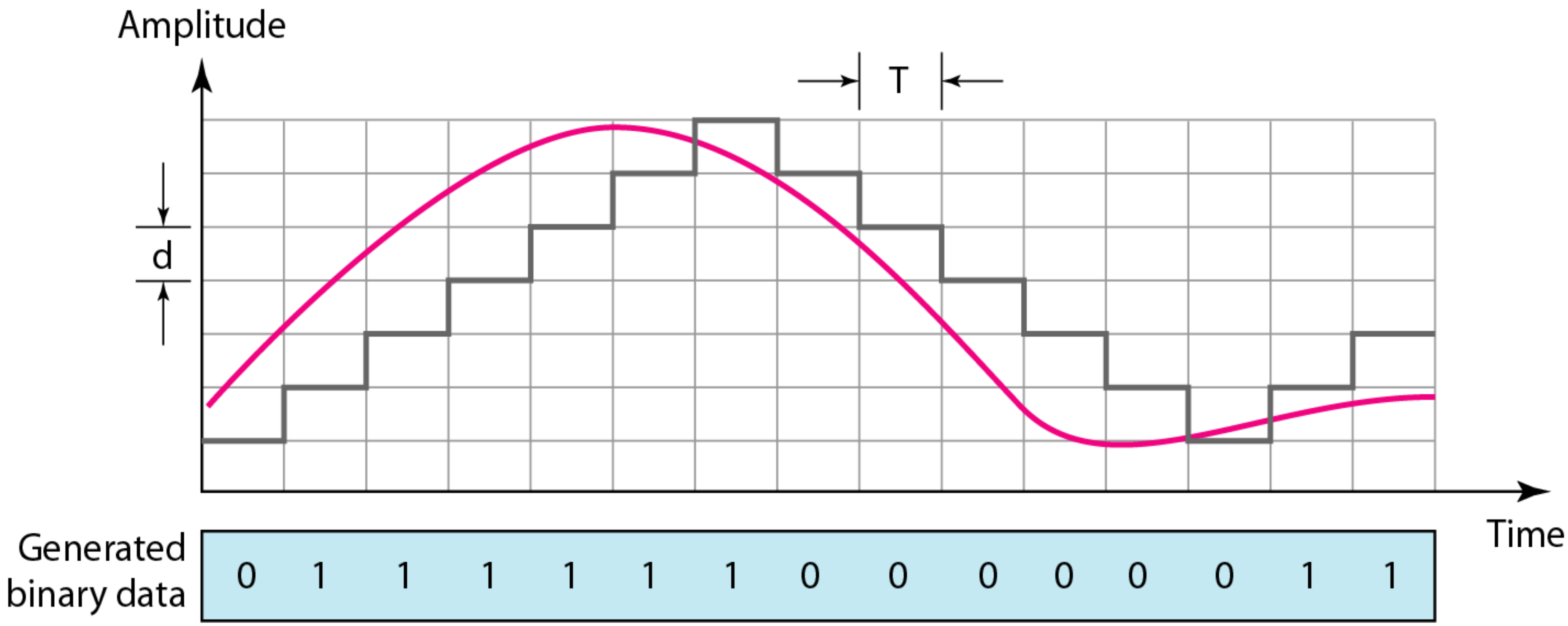
Delta Modulation

- In Delta Modulation, only one bit is transmitted per sample
- That bit is a one if the current sample is more positive than the previous sample, and a zero if it is more negative
- Since so little information is transmitted, delta modulation requires higher sampling rates than PCM for equal quality of reproduction

Delta Modulation

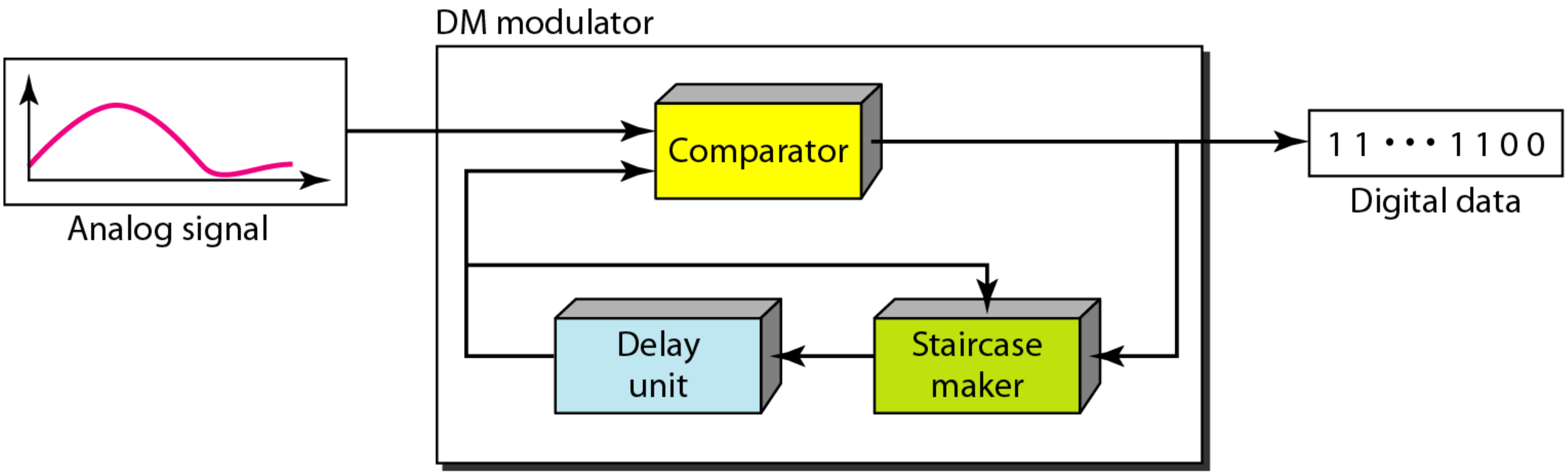
- This scheme sends only the difference between pulses, if the pulse at time t_{n+1} is higher in amplitude value than the pulse at time t_n , then a single bit, say a “1”, is used to indicate the positive value.
- If the pulse is lower in value, resulting in a negative value, a “0” is used.
- This scheme works well for small changes in signal values between samples.
- If changes in amplitude are large, this will result in large errors.

Delta Modulation

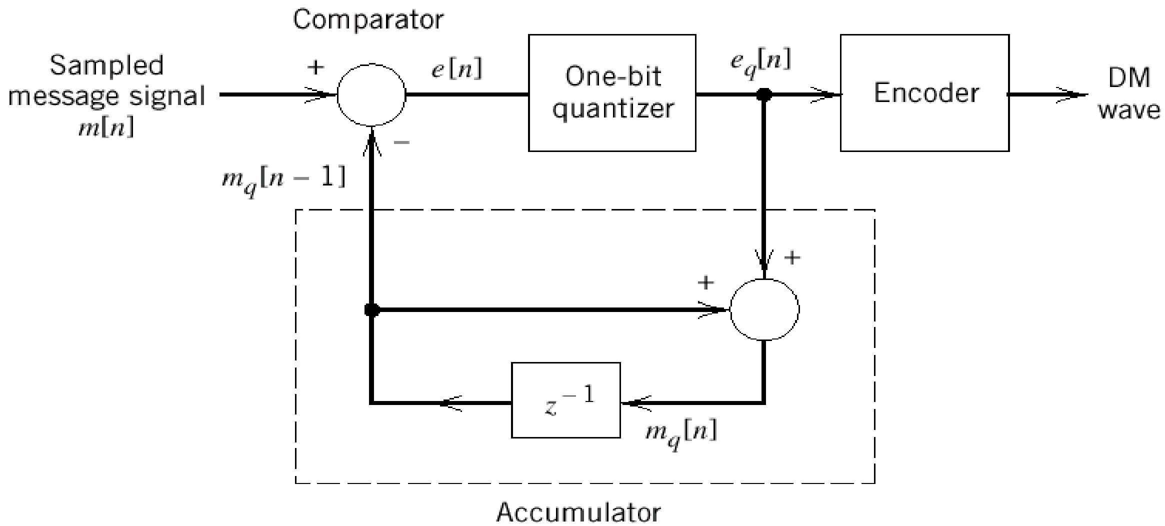


The process of delta modulation

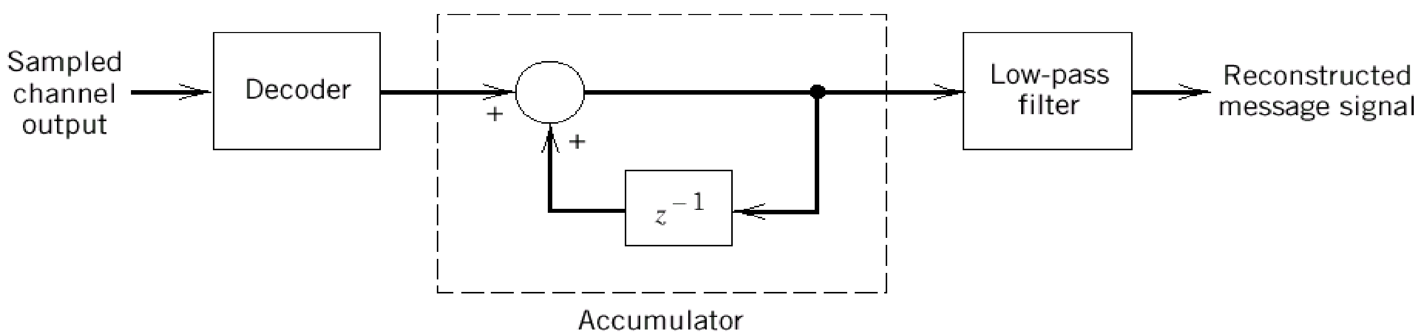
Components of Delta Modulation



DM system.



(a) Transmitter.

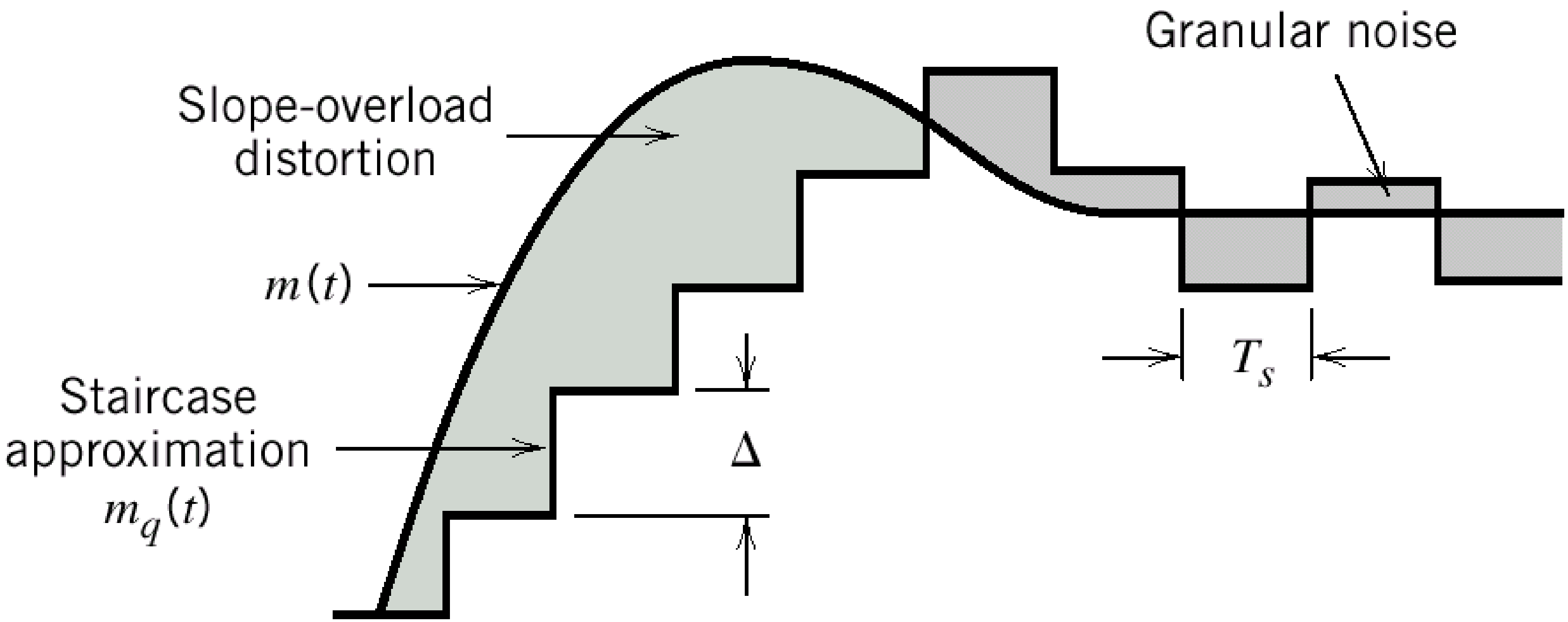


(b) Receiver

Distortions in DM system

1. If the slope of analog signal is much higher than that of approximated digital signal over long duration, than this difference is called **Slope overload distortion**.
2. The difference between quantized signal and original signal is called as **Granular noise.** It is similar to quantisation noise.

Quantization errors



Two types of quantization errors: **Slope overload distortion** and **granular noise**

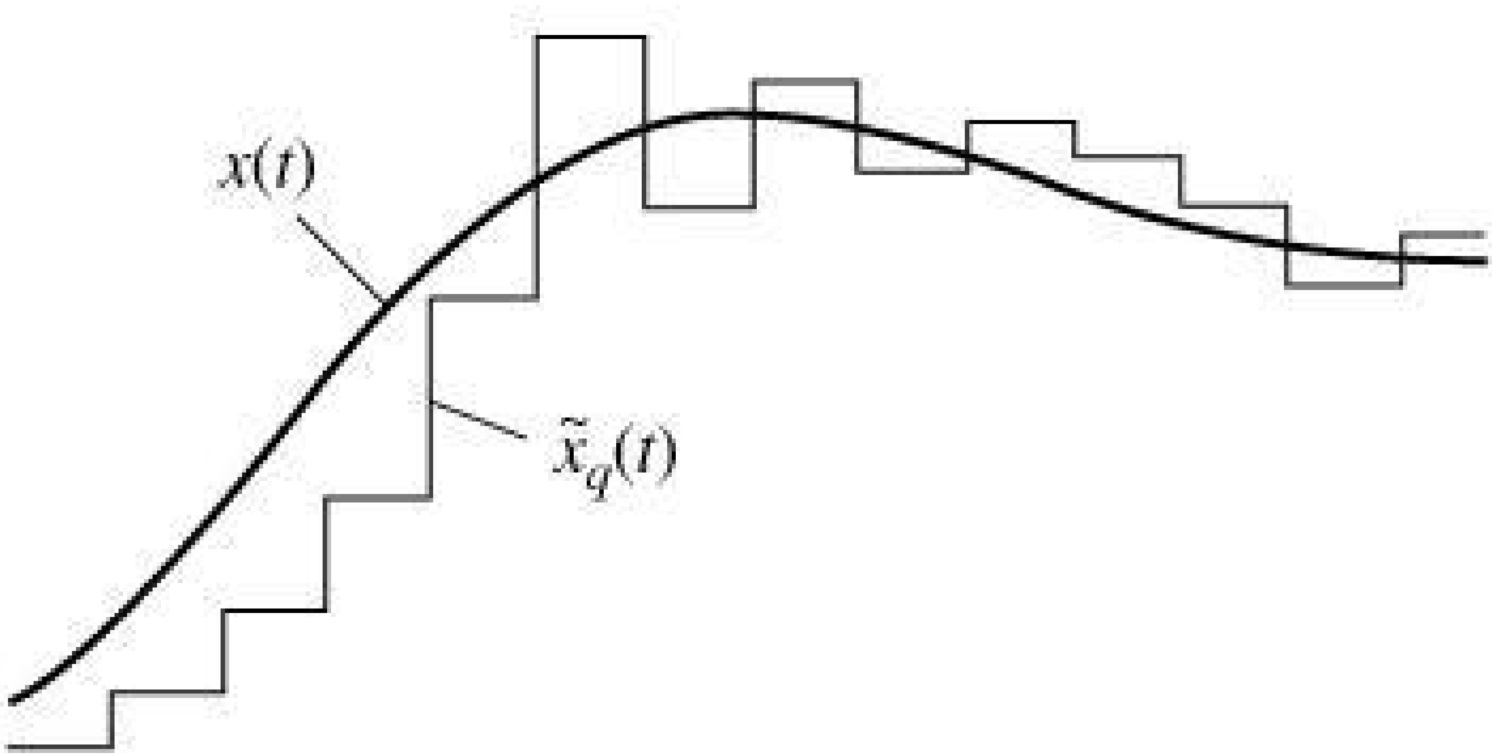
Distortions in DM System

- Granular noise occurs when step size ▲ is large relative to local slope m(t).
- There is a further modification in this system, in which step size is not fixed.
- That scheme is known as **Adaptive Delta Modulation**

Adaptive Delta Modulation

- A better performance can be achieved if the value of ▲is not fixed.
- The value of ▲changes according to the amplitude of the analog signal.
- It has wide dynamic range due to variable step size.
- Also better utilisation of bandwidth has compared to delta modulation.
- Improvement in signal to noise ratio

Adaptive Delta Modulation



Differential Pulse Code Modulation (DPCM)

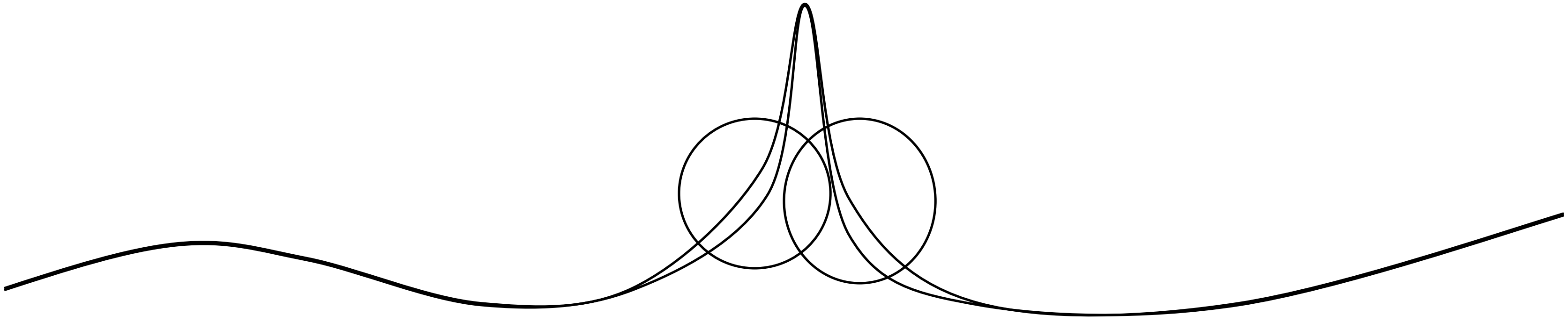
- What if we look at sample differences, not the samples themselves?
  - $d_t = x_t - x_{t-1}$
  - Differences tend to be smaller
    - Use 4 bits instead of 12, maybe?

Differential Pulse Code Modulation (DPCM)

- Changes between adjacent samples small
- Send value, then relative changes
  - value uses full bits, changes use fewer bits
  - E.g., 220, 218, 221, 219, 220, 221, 222, 218,.. (all values between 218 and 222)
  - Difference sequence sent: 220, +2, -3, 2, -1, -1, -1, +4....
  - Result: originally for encoding sequence 0..255 numbers need 8 bits;
  - Difference coding: need only 3 bits

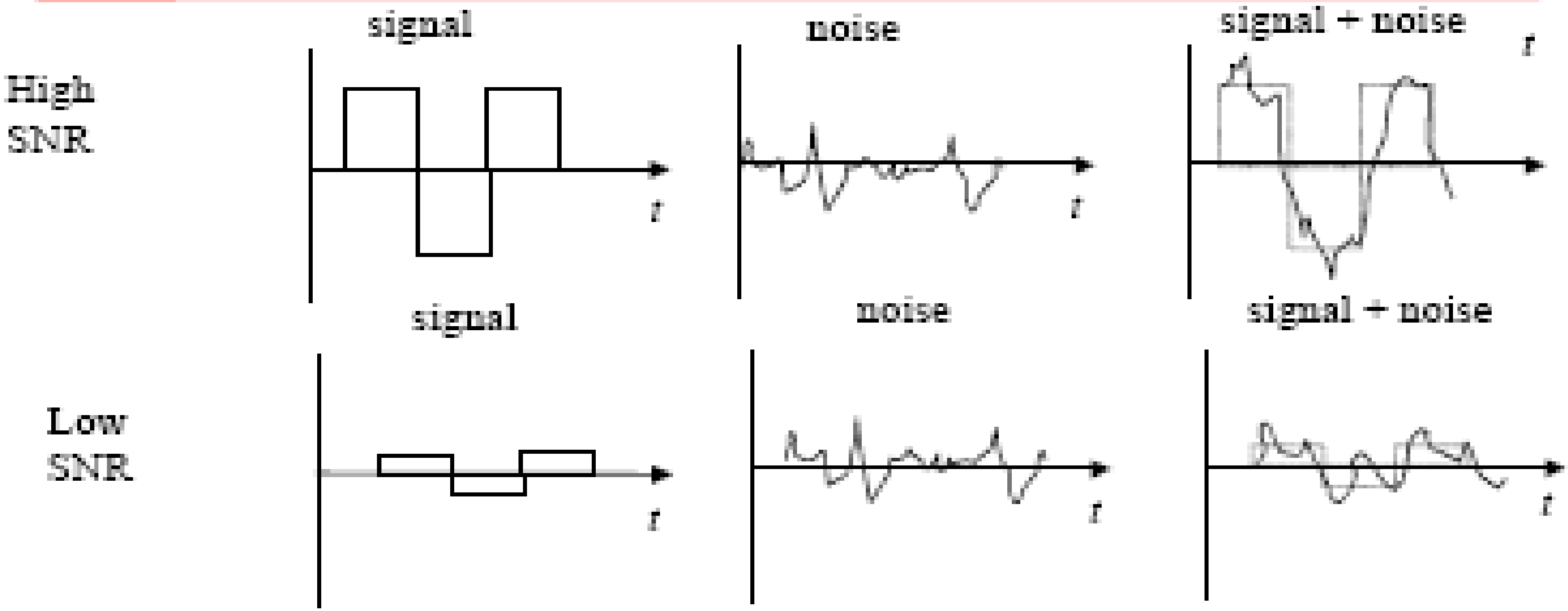
Adaptive Differential Pulse Code Modulation (ADPCM)

- Adaptive similar to DPCM, but adjusts the width of the quantization steps
- Encode difference in 4 bits, but vary the mapping of bits to difference dynamically
  - If rapid change, use large differences
  - If slow change, use small differences



Signal-to-Noise Ratio

(metric to quantify quality of digital audio)



$$\text{SNR} = \frac{\text{Average Signal Power}}{\text{Average Noise Power}}$$

$$\text{SNR (dB)} = 10 \log_{10} \text{SNR}$$

Signal To Noise Ratio

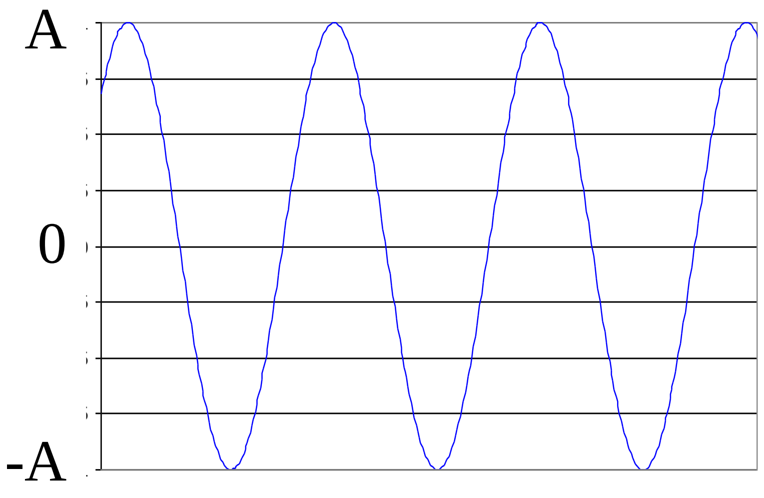
- Measures strength of signal to noise

SNR (in DB)=

$$10\log_{10}\left(\frac{Signal\,energy}{Noise\,energy}\right)$$

- Given sound form with amplitude in [-A, A]

- Signal energy = $\dfrac{A^2}{2}$



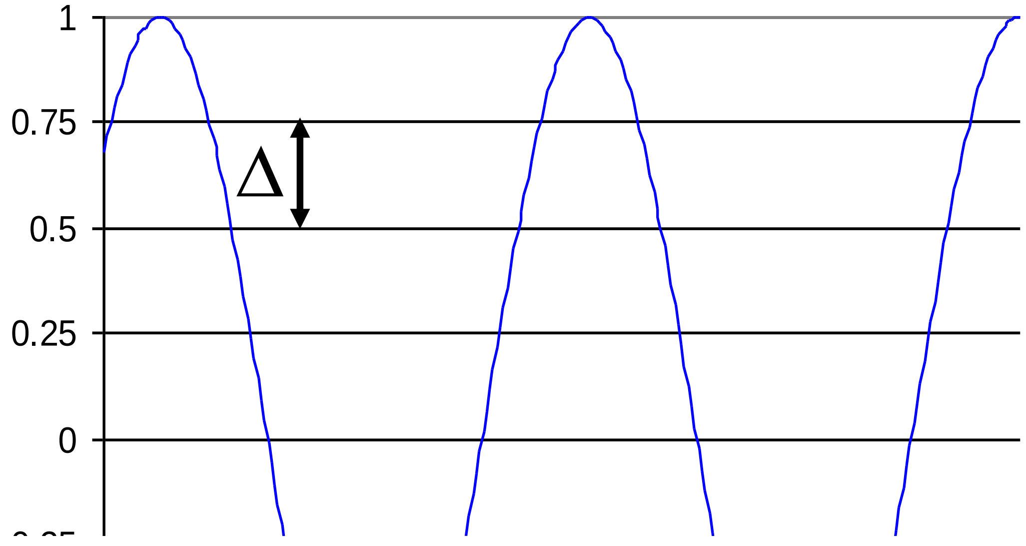
Quantization Error

- Difference between actual and sampled value
  - amplitude between [-A, A]
  - quantization levels = N

$$\Delta = \frac{2A}{N}$$

- e.g., if A = 1,

  N = 8, Δ = 1/4



Examples

- 2.1 A sinusoidal input wave of 3kHz is to be sampled at the lowest rate for transmission as pulses. Calculate the minimum sampling frequency required, so that all components of the wave can be reconstructed at the receiver.
- 2.2 The PCM sampled are encoded into 4-bits system. If the minimum sampling rate used is 8kHz, calculate

a) the frequency of the information signal
b) the quantization level.
c) the transmission rate
d) The transmission bandwidth

Thank You